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Project Title	Unsupervised Financial Anomaly Detection via Isolation Forest from First Principles
Week	Week 7

WEEK- 7 Technical Report

Task Name

Unsupervised Financial Anomaly Detection via Isolation Forest from First Principles

Abstract

This project focuses on detecting unusual financial transactions using the Isolation Forest algorithm implemented from scratch. The model isolates anomalies by recursively partitioning transaction data using randomly selected features and split values. Transactions that are isolated in fewer steps are considered anomalies. The system constructs multiple isolation trees and computes anomaly scores mathematically to identify suspicious financial activities without requiring labeled data.

Introduction

Financial institutions process millions of transactions daily. Detecting fraudulent or abnormal transactions is important for reducing financial risks. Traditional supervised methods require labeled datasets, which may not always be available. Isolation Forest is an unsupervised machine learning technique that identifies anomalies by isolating data points through random binary partitioning. Anomalous transactions are easier to isolate and therefore have shorter path lengths in isolation trees.

Objectives

- Detect anomalous financial transactions without labeled data.
- Implement Isolation Forest from first principles.
- Build multiple isolation trees using recursive binary partitioning.
- Calculate anomaly scores based on average path lengths.

- Identify suspicious transactions for further investigation.

Methodology

1. Collect financial transaction data.
2. Select transaction attributes (Amount, Balance, Time, Frequency, etc.).
3. Build isolation trees using random features and split values.
4. Measure path length for each transaction.
5. Compute average path length across all trees.
6. Calculate anomaly scores.
7. Flag transactions with high anomaly scores as anomalies.

Mathematical Formula

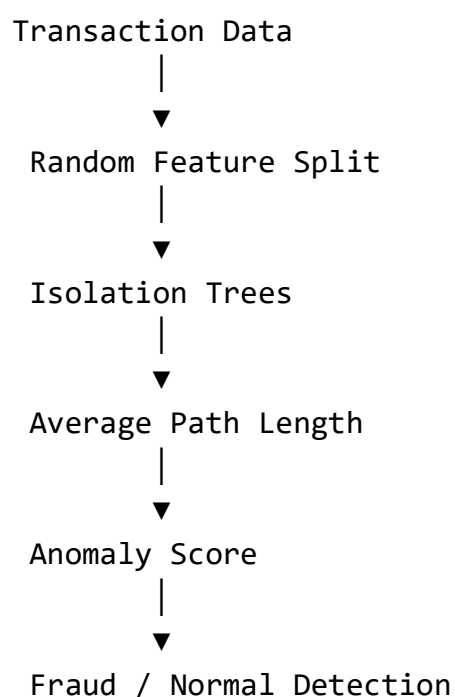
Anomaly Score:

$$S(x,n) = 2 - \frac{E(h(x))}{c(n)}$$

Where:

- $S(x,n)$ = Anomaly score
- $E(h(x))$ = Average path length
- $c(n)$ = Average path length of unsuccessful search in Binary Search Tree
- n = Number of samples

Simple Diagram



Conclusion

The Isolation Forest algorithm efficiently detects unusual financial transactions by isolating anomalies through random partitioning. Transactions with shorter path lengths receive higher anomaly scores and are classified as suspicious. This approach works effectively without labeled data and is suitable for large-scale financial monitoring systems.